

Goal. Explore methods of determining whether series converge or diverge.

0. State the direct comparison in your own words. What conditions do you need to check in order to use this test?

You can also use the following test.

Limit comparison test

Suppose $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms ($a_n \geq 0$ and $b_n \geq 0$) such that $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L$ for some number L .

(a) If $0 < L < \infty$, then the series either both converge or both diverge.

(b) If $L = \infty$ and $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} b_n$ converges.

(c) If $L = 0$ and $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.

1. Use any of the following tests to determine whether the following series converge or diverge: n th term divergence test, Integral test, Comparison test, Limit comparison test. Be sure to verify the hypotheses, state your conclusion, and say which test you used.

(a)
$$\sum_{n=2}^{\infty} \frac{1}{\sqrt{n^4 + n^3}}$$

(b)
$$\sum_{n=1}^{\infty} \frac{1}{4 + 5^n}$$

(c)
$$\sum_{n=1}^{\infty} \frac{n^3 - 5n + 1}{7n^5 + 2}$$

$$(d) \sum_{n=1}^{\infty} \frac{e^{-n}}{n + \cos^2(n)}$$

$$(e) \sum_{n=2}^{\infty} \frac{4n^2 + n}{\sqrt[3]{n^7 + n^3}}$$

$$(f) \sum_{k=2}^{\infty} \sqrt{\frac{k}{k^3 + 1}}$$